Day 103 — TNPSC Group 2A Study Material

Part A – General Studies: Current Affairs in Geography of India (9): India's Port Capacity & Maritime Infrastructure (2026)

READING MATERIAL

Covers India's maritime geography fundamentals and 2026 port-infrastructure milestones.

Q: What are India's coastline length and Exclusive Economic Zone (EEZ) area, and what share of GDP does the maritime sector contribute?

A: Coastline: 11,098 km. EEZ: 2.3 million sq km. Maritime sector contributes about 4% of GDP, with ports handling 95% of India's trade volume.

Q: What is India's total port cargo-handling capacity as of 2026, and which port is India's largest commercial port?

A: 2,762 million tonnes per annum (MTPA) total capacity. Mundra Port is India's largest commercial port — the first Indian port to handle 200 million tonnes (MT) of cargo in a single year.

Q: What is distinctive about Vizhinjam Port, Kerala?

A: It is India's first fully automated deep-water container terminal, designed to handle ultra-large 'megamax' ships, with automated quay cranes and rail-mounted gantries.

Q: How has average port turnaround time changed from 2010-11 to FY26?

A: Fallen from 127 hours (2010-11) to 49.32 hours (FY26) — a major efficiency improvement.

Q: What technology milestone did an Indian major port achieve in March 2026?

A: Became the first port in India to implement Digital Twin technology, enabling real-time monitoring and predictive analysis of port operations.

MOCK TEST (WITH ANSWERS)

1. India's coastline length is approximately:

A. 7,516 km

B. 9,098 km

C. 11,098 km ✓

D. 13,098 km

Explanation: 11,098 km. [medium, web-research]

2. Which port became the first in India to handle 200 million tonnes of cargo in a single year?

A. Jawaharlal Nehru Port

B. Mundra Port ✓

C. Vizhinjam Port

D. Kandla Port

Explanation: Mundra Port. [medium, web-research]

3. Vizhinjam Port (Kerala) is notable as India's first:

A. Inland river port

B. Fully automated deep-water container terminal ✓

C. Floating port

D. Naval-only port

Explanation: Fully automated deep-water container terminal. [medium, web-research]

4. India's total port cargo-handling capacity as of 2026 is approximately:

A. 1,762 MTPA

B. 2,262 MTPA

C. 2,762 MTPA ✓

D. 3,262 MTPA

Explanation: 2,762 MTPA. [hard, web-research]

5. Average port turnaround time in India has fallen from 127 hours (2010-11) to approximately how much in FY26?

A. 29.32 hours

B. 39.32 hours

C. 49.32 hours ✓

D. 69.32 hours

Explanation: 49.32 hours. [hard, web-research]

Part B – Aptitude & Mental Ability: Compound Interest

READING MATERIAL

Compound Interest is 1 of 10 named Aptitude sub-skills. Real exam evidence shows TNPSC tests the standard CI formula, a useful SI-vs-CI shortcut, AND disguises CI as 'successive percentage change' word problems (depreciation, year-over-year growth) — recognize all three shapes.

Q: Formula: What is the Compound Interest formula, and the shortcut for the DIFFERENCE between SI and CI over 2 years?

A: Amount = P(1+R/100)^T, so CI = Amount − P. For exactly 2 years, there's a shortcut: CI − SI = P×(R/100)². This avoids computing CI and SI separately when you only need their difference.

Q: Real exam (2022): Find the difference between SI and CI for 2 years on a principal of Rs.5,000 at 8% per annum.

A: Using the shortcut: difference = P×(R/100)² = 5000×(0.08)² = 5000×0.0064 = Rs.32. (Full check: SI = 5000×8×2/100=800. CI = 5000×(1.08²−1) = 5000×0.1664 = 832. Difference = 832−800 = 32 ✓.)

Q: Real exam (2022): Find the compound interest on Rs.5,000 at 12% per annum for 2 years, compounded annually.

A: CI = P×((1+R/100)^T − 1) = 5000×(1.12² − 1) = 5000×(1.2544−1) = 5000×0.2544 = Rs.1,272.

Q: Real exam (2025): A sum becomes Rs.13,380 after 3 years and Rs.20,070 after 6 years on compound interest. Find the original sum.

A: The amount after 6 years equals the amount after 3 years multiplied by the SAME growth factor again (since compounding is multiplicative): (1+R/100)³ = 20070/13380 = 1.5. The sum itself: P = Amount-after-3-years ÷ (1+R/100)³ = 13380 ÷ 1.5 = Rs.8,920.

Q: Real exam (2024): A motorcycle's value 2 years ago was Rs.80,000. It depreciates 4% in the first year, then increases 4% in the second year. Find the present value.

A: Apply each year's change as its own multiplier, in sequence — this is compound interest with a changing/mixed rate: 80,000 × 0.96 (first year, −4%) × 1.04 (second year, +4%) = 80,000 × 0.9984 = Rs.79,872. Note the result is slightly LESS than the original Rs.80,000 — a decrease followed by an equal-percentage increase never fully cancels out, since the increase applies to a smaller base.

MOCK TEST (WITH ANSWERS)

1. Find the difference between SI and CI for 2 years on a principal of Rs.8,000 at 5% per annum.

A. Rs.18

B. Rs.20 ✓

C. Rs.22

D. Rs.25

Explanation: Difference = P×(R/100)² = 8000×(0.05)² = 8000×0.0025 = Rs.20. [medium, authored]

2. Find the compound interest on Rs.10,000 at 10% per annum for 2 years, compounded annually.

A. Rs.2,000

B. Rs.2,100 ✓

C. Rs.2,200

D. Rs.1,900

Explanation: CI = 10000×(1.10²−1) = 10000×0.21 = Rs.2,100. [easy, authored]

3. A sum becomes Rs.16,000 after 2 years and Rs.19,360 after 4 years on compound interest. Find the original sum.

A. Rs.12,500

B. Rs.13,000

C. Rs.13,223 ✓

D. Rs.14,000

Explanation: (1+R/100)² = 19360/16000 = 1.21. P = 16000/1.21 ≈ Rs.13,223. [hard, authored]

4. A machine valued at Rs.50,000 depreciates by 10% in the first year and a further 10% in the second year. Find its value after 2 years.

A. Rs.40,000

B. Rs.40,500 ✓

C. Rs.41,000

D. Rs.39,500

Explanation: 50,000 × 0.90 × 0.90 = 50,000 × 0.81 = Rs.40,500. [medium, authored]

5. Find the compound interest on Rs.20,000 at 15% per annum for 2 years, compounded annually.

A. Rs.6,000

B. Rs.6,450 ✓

C. Rs.6,200

D. Rs.6,500

Explanation: CI = $20000\times(1.15^2-1) = 20000\times0.3225$ = Rs.6,450. [medium, authored]

Part C – General English: Poem: "Nature, the Gentlest Mother" — Emily Dickinson

READING MATERIAL

A poem personifying Nature as a patient, caring mother who gently disciplines her 'children' (plants, animals, the natural world). Real coaching-site evidence tests personification examples and the poem's caring/patient tone heavily.

Q: Who wrote "Nature, the Gentlest Mother", and what is its central theme?

A: Emily Dickinson. Central theme: Nature personified as an infinitely patient, caring mother — looking after all her 'children' (plants, animals, the natural world) with gentle, unbiased compassion, regardless of their shortcomings.

Q: How is Nature's temperament described in the poem?

A: Gentle and patient — not harsh, angry, cold, or distant.

Q: Identify the figure of speech: "Nature, the gentlest mother."

A: Personification — Nature (an abstract/non-human concept) is directly given the human role and qualities of a 'mother'. This is the poem's primary, sustained device throughout.

Q: What does "Her admonition mild" suggest about how Nature corrects wrongdoing?

A: Nature gently corrects — not through scolding, anger, or punishment, but mild, patient guidance.

Q: How does Nature 'discipline' her creatures, per the poem's examples?

A: By 'restraining' an overly active ('Rampant') squirrel, or muffling a 'too impetuous' bird — Nature's discipline is gentle moderation, not harsh punishment, shown through small natural behaviours rather than dramatic intervention.

Q: Who is said to hear Nature's 'conversation' on a summer afternoon, and what figure of speech appears in "Her golden finger on her lip"?

A: Travellers hear it. "Her golden finger on her lip" is Personification — Nature given the human gesture of placing a finger to the lips (a 'hush' gesture), continuing the maternal personification.

MOCK TEST (WITH ANSWERS)

1. "Nature, the Gentlest Mother" was written by:

A. Maya Angelou

B. Robert Frost

C. Emily Dickinson ✓

D. Sarojini Naidu

Explanation: Emily Dickinson. [easy, web-research]

2. How is Nature described in the poem?

A. Harsh and angry

B. Gentle and patient ✓

C. Cold and distant

D. Loud and wild

Explanation: Gentle and patient. [easy, web-research]

3. "Nature, the gentlest mother" is an example of which figure of speech?

A. Simile

B. Metaphor

C. Personification ✓

D. Alliteration

Explanation: Personification — Nature given the role of a mother. [easy, web-research]

4. "Her admonition mild" suggests that Nature:

A. Scolds angrily

B. Ignores wrongdoing entirely

C. Gently corrects ✓

D. Punishes severely

Explanation: Gently corrects. [medium, web-research]

5. According to the poem, who hears Nature's 'conversation' on a summer afternoon?

A. Teachers

B. Scientists

C. Travellers ✓

D. Soldiers

Explanation: Travellers. [medium, web-research]

6. "Her golden finger on her lip" uses which figure of speech?

A. Simile

B. Metaphor

C. Personification ✓

D. Alliteration

Explanation: Personification — a human 'hush' gesture given to Nature. [medium, web-research]